

The Random Subspace Method for Forecast Combination

Theoretical Model

Boris Kozyrev

Job Market Paper

June 23, 2026

Roadmap

- 1 Motivation
- 2 Setup and Loss Function
- 3 RSM: Definition and the Bagging Bridge
- 4 RSM as a Unifier
- 5 Common-Loading Structure
- 6 RSM Risk and the Optimal k^*

The forecast-combination puzzle

A long-standing puzzle

The *equal-weighted mean* of individual forecasts typically **outperforms** the estimated minimum-variance “optimal” combination — even though the latter solves a simple constrained OLS problem. (Bates and Granger, 1969; Granger and Ramanathan, 1984)

Two standard explanations:

- **Estimation error.** Optimal weights must be estimated; with m forecasts and short samples T , the noise overwhelms the gains.
- **Small population gains.** Elliott and Liao (2025): even *before* any finite-sample noise, the population gain $Q^{\text{ave}} - Q^{\text{opt}}$ is often tiny under empirically plausible covariance structures (probably, not true in the endogenous case).

This paper

Both forces point to using *fewer* forecasts. The Random Subspace Method (RSM) tries to answer how many of them are needed.

The inflation factor

Estimating weights inflates the population loss. For the full m -weight optimal combination (constrained OLS):

$$I_m(T) = \frac{T-1}{T-m} \approx 1 + \frac{m-1}{T}.$$

Elliott and Liao (2025) call $(m-1)/T$ the *best case* estimation cost. Optimal combination is worth it only if

$$Q^{\text{ave}} - Q^{\text{opt}} > \frac{m-1}{T} (\sigma_\varepsilon^2 + Q^{\text{opt}}),$$

which needs $T \gg m$ to hold (cost negligible); but in reality T is of moderate size, so it often **fails**.

The RSM departure

Use a **subset** of size $k \ll m$, with inflation $\frac{T-1}{T-k} \ll \frac{T-1}{T-m}$, i.e. cost $\frac{k-1}{T} \ll \frac{m-1}{T}$.

A window of *net* gains opens even when full OLS is not worth estimating.

What RSM adds over the literature

Elliott–Liao When gains are largest, the optimum is a fixed subset with equal within-subset weights. But: *which* forecasts? *how large* the subset? — left open.

RSM closes both gaps and goes further:

- ➊ **Selection via randomisation** — replaces an intractable $\binom{m}{k}$ search with uniform random draws + averaging.
- ➋ **In population**, RSM contains both equal-weighting ($k = 1$) and optimal weights $k = m$.
- ➌ **A theory for k^*** — the (potential) optimal subset size, which neither Elliott and Liao (2025) nor Chen and Maung (2023) provide.

The central question

In a panel of m forecasters, *how many* should we randomly subsample, and *how* should we combine them, to minimise out-of-sample loss?

Setup

m individual forecasts $f_t = (f_{1t}, \dots, f_{mt})'$ of a scalar target $y_t = \mu_t + \epsilon_t$. Forecast errors: $u_t = f_t - \mu_t \mathbf{1}_m$.

Assumption (Target innovation)

$$\mathbb{E}(\epsilon_t) = 0, \quad \text{Var}(\epsilon_t) = \sigma_\epsilon^2 > 0, \quad \mathbb{E}(\epsilon_t u_{it}) = 0 \quad \forall i.$$

Assumption (Forecast-error covariance)

$$\mathbb{E}(u_t) = 0, \quad \text{Var}(u_t) = \Sigma_{e,m} \succ 0.$$

Following Elliott and Liao (2025):

$$\begin{pmatrix} y_t \\ f_t \end{pmatrix} \sim \left(\begin{pmatrix} \mu_t \\ \mu_t \end{pmatrix}, \begin{pmatrix} \sigma_\epsilon^2 & 0 \\ 0 & \Sigma_{e,m} \end{pmatrix} \right).$$

$\Sigma_{e,m}$ summarises cross-forecast dispersion (diagonal: marginal error variances; off-diagonal: co-movement). σ_ϵ^2 is the *irreducible noise* shared by every combination.

The loss function

A combined forecast $w'_m f_t$ with $w'_m \iota_m = 1$ gives error $w'_m u_t - \epsilon_t$ and MSE

$$L = \underbrace{(1 - w'_m \iota_m)^2 \mu_t^2}_{\text{Bias}} + \underbrace{\sigma_\epsilon^2 + w'_m \Sigma_{e,m} w_m}_{\text{Variance}}.$$

Under the sum-to-one constraint the combination is **unbiased**, so the bias term vanishes:

$$L = \sigma_\epsilon^2 + w'_m \Sigma_{e,m} w_m.$$

- Comparison across methods reduces to minimising $w'_m \Sigma_{e,m} w_m$.
- σ_ϵ^2 is common to all combinations — but it *enters* the loss and gets inflated by estimation error, so (unlike Boot and Nibbering, 2019) we keep it.

The two benchmarks

Equal weighting

$$w^{\text{ave}} = m^{-1} \iota_m$$
$$L^{\text{ave}} = \sigma_\varepsilon^2 + \underbrace{m^{-2} \iota'_m \Sigma_{e,m} \iota_m}_{Q^{\text{ave}}}$$

No weights estimated $\Rightarrow$ *no* estimation penalty.

Optimal weighting

$$w^{\text{opt}} = \frac{\Sigma_{e,m}^{-1} \iota_m}{\iota'_m \Sigma_{e,m}^{-1} \iota_m}$$

$$L^{\text{opt}} \approx (\sigma_\varepsilon^2 + Q^{\text{opt}}) \frac{T-1}{T-m}$$

with $Q^{\text{opt}} = (\iota'_m \Sigma_{e,m}^{-1} \iota_m)^{-1}$.

Inflation factor

We use the **exact** out-of-sample inflation $I(T, k) = \frac{T-1}{T-k}$ for $k-1$ estimated coefficients. Elliott and Liao (2025) use its first-order approximation $1 + \frac{m-1}{T}$ (valid for $m \ll T$).

RSM: subset weights

Draw a subset $S \subset \{1, \dots, m\}$, $|S| = k$, uniformly without replacement from $\binom{m}{k}$ subsets. Let $\Sigma_{e,k,S}$ be the $k \times k$ principal submatrix.

Within-subset optimal weights

$$w_S = \frac{\Sigma_{e,k,S}^{-1} \boldsymbol{\iota}_k}{\boldsymbol{\iota}_k' \Sigma_{e,k,S}^{-1} \boldsymbol{\iota}_k}.$$

Embed into $\mathbb{R}^m$ via the zero-padding map P_S :

$$v_S = P_S w_S, \quad \sum_{i=1}^m v_{S,i} = 1.$$

Zero-padding preserves the quadratic form:

$$v_S' \Sigma_{e,m} v_S = w_S' \Sigma_{e,k,S} w_S = Q(S), \quad Q(S) = (\boldsymbol{\iota}_k' \Sigma_{e,k,S}^{-1} \boldsymbol{\iota}_k)^{-1}.$$

Three population risks

Definition (Average-subset RSM)

Single random draw, expected loss over all subsets:

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S' \Sigma_{e,m} v_S] = \mathbb{E}_S[Q(S)].$$

Definition (Finite-bagged RSM)

G independent draws, $\hat{v}_{k,G} = \frac{1}{G} \sum_g v_{S_g}$, then

$$Q_k^{\text{bag},G} = \mathbb{E}[\hat{v}_{k,G}' \Sigma_{e,m} \hat{v}_{k,G}].$$

Loss of the aggregated forecast — not the average of losses.

Definition (Infinite-bagged RSM)

$\bar{v}_k = \mathbb{E}_S[v_S]$ as $G \rightarrow \infty$, with deterministic risk

Convexity: bagging never hurts

Since $\Sigma_{e,m} \succ 0$, the map $v \mapsto v' \Sigma_{e,m} v$ is strictly convex. Jensen's inequality gives

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S' \Sigma_{e,m} v_S] \geq (\mathbb{E}_S[v_S])' \Sigma_{e,m} (\mathbb{E}_S[v_S]) = Q_k^{\text{bag}, \infty}.$$

- Equality *iff* v_S is constant across subsets — which happens only when $k = m$.
- Otherwise the inequality is **strict**: infinite bagging strictly reduces population risk relative to a single draw.

Takeaway

Averaging weight vectors (bagging) is a free variance reduction — no distributional assumptions needed beyond $\Sigma_{e,m} \succ 0$.

The finite-bagging bridge

Proposition (Finite-bagging bridge)

For independent subset draws,

$$Q_k^{\text{bag}, G} = Q_k^{\text{bag}, \infty} + \frac{1}{G} (Q_k^{\text{sub}} - Q_k^{\text{bag}, \infty}),$$

hence

$$Q_k^{\text{bag}, \infty} \leq Q_k^{\text{bag}, G} \leq Q_k^{\text{sub}}.$$

- Holds for *any* distribution of $\{f_t\}$.
- Increasing G **monotonically** lowers the finite-bagged risk toward $Q_k^{\text{bag}, \infty}$.
- Special cases: $Q_k^{\text{bag}, G} = Q_k^{\text{sub}}$ at $G = 1$; $Q_k^{\text{bag}, G} \rightarrow Q_k^{\text{bag}, \infty}$ as $G \rightarrow \infty$.
- Finite bagging is the practitioner's object: intermediate between the two limits, arbitrarily close to $Q_k^{\text{bag}, \infty}$ for large G .

What can we say with *no* structure?

What survives if we know *nothing* about $\Sigma_{e,m}$ beyond $\Sigma_{e,m} \succ 0$ — no common loading, no distribution of $\{f_t\}$?

Two-sided sandwich

$$Q^{\text{opt}} \leq Q_k^{\text{bag},\infty} \leq Q_k^{\text{sub}}.$$

- **Lower bound (feasibility).** $\bar{v}_k = \mathbb{E}_S[v_S]$ is an admissible sum-to-one combination ($\bar{v}'_k \mathbf{1}_m = 1$). Since w^{opt} *minimises* $w' \Sigma_{e,m} w$ over all such w , $Q_k^{\text{bag},\infty} = \bar{v}'_k \Sigma_{e,m} \bar{v}_k \geq Q^{\text{opt}}$.
- **Upper bound (Jensen).** Convexity of $v \mapsto v' \Sigma_{e,m} v$ gives $Q_k^{\text{bag},\infty} \leq Q_k^{\text{sub}}$.
- **Anchored at the benchmarks.** $Q_1^{\text{bag},\infty} = Q^{\text{ave}}$, $Q_m^{\text{bag},\infty} = Q^{\text{opt}}$.

Takeaway

In *population* bagged RSM never beats the (infeasible) optimum and never does worse than a single draw. Any out-of-sample *win over optimal* comes from the smaller inflation factor $\frac{T-1}{T-k}$ — not from population risk.

Boundary cases: $k = 1$ and $k = m$

Singletons ($k = 1$): every draw assigns full weight to one forecast, $v_{\{i\},i} = 1$. Drawn with prob. $1/m$:

$$\bar{v}_{1,i} = \mathbb{E}_S[v_{S,i}] = \frac{1}{m} \Rightarrow \bar{v}_1 = \frac{1}{m} \boldsymbol{\iota}_m = w^{\text{ave}}.$$

Full set ($k = m$): only one subset, no randomness:

$$\bar{v}_m = \frac{\sum_{e,m}^{-1} \boldsymbol{\iota}_m}{\boldsymbol{\iota}_m' \sum_{e,m}^{-1} \boldsymbol{\iota}_m} = w^{\text{opt}}.$$

Proposition (RSM as a unifier)

Both benchmarks are special cases of RSM: $k = 1$ collapses (in population) to equal weighting; $k = m$ yields optimal weighting. RSM interpolates as k grows.

Homogeneous forecasts $\Rightarrow$ small k optimal; $T \gg m \Rightarrow k = m$. Most real panels are heterogeneous with short $T \Rightarrow$ an interior $1 < k < m$ wins.

Global vs. conditional weights

With uniform size- k subsets, $P(i \in S) = k/m$, so the global weight is

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = P(i \in S) \mathbb{E}_S[w_{S,i} \mid i \in S] = \frac{k}{m} \mathbb{E}_S[w_{S,i} \mid i \in S].$$

Proposition (Global–conditional correspondence)

$$\bar{v}_{k,i} > \frac{1}{m} \iff \mathbb{E}_S[w_{S,i} \mid i \in S] > \frac{1}{k}.$$

- A forecast earns above-average *global* weight iff it earns above-average weight *within the subsets it appears in*.
- “Global importance is achieved locally.”
- Propositions on this slide and the last require *only* $\Sigma_{e,m} \succ 0$ — exact identities, no structural assumptions.

Common-loading heteroskedastic errors

Assumption (Forecast errors)

$$u_t = \lambda F_t \boldsymbol{\iota}_m + \xi_t, \quad \text{Var}(\lambda F_t) = q > 0, \quad \text{Var}(\xi_t) = D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2).$$

A common shock (same loading λ) plus idiosyncratic noise:

$$\Sigma_{e,m} = q \boldsymbol{\iota}_m \boldsymbol{\iota}_m' + D, \quad \Sigma_{e,S} = q \boldsymbol{\iota}_k \boldsymbol{\iota}_k' + D_S.$$

Definition (Precision and precision sums)

$$\tau_i = \sigma_i^{-2}, \quad A_S = \sum_{i \in S} \tau_i, \quad A_m = \sum_{i=1}^m \tau_i.$$

Subset risk under common loading

Proposition (Subset risk)

For any subset S of size k ,

$$Q(S) = q + \frac{1}{A_S}, \quad w_{i,S} = \frac{\tau_i}{A_S} \quad (i \in S).$$

- Loss = irreducible common component q + inverse sum of drawn precisions.
- $Q(S)$ is strictly decreasing in k along nested subsets ($\tau_i > 0$); minimised at $k = m$ ($\rightarrow Q^{\text{opt}}$).
- **But** the inflation factor $I(T, m)$ may stop full optimal from winning in practice.
- Optimal subset weights depend only on the precisions τ — *not* on q .

Optimal vs. equal weighting: the B – H decomposition

Optimal weight $w_i^{\text{opt}} = \tau_i / A_m$, loss $Q^{\text{opt}} = q + 1/A_m$. Equal weighting ignores heterogeneity: $Q^{\text{ave}} = q + (\sum_i \tau_i^{-1}) / m^2$.

Define $B = 1/\bar{\tau}$ (reciprocal mean precision; idiosyncratic *scale*) and $H = \overline{\sigma^2} \bar{\tau}$ (arithmetic/harmonic variance ratio; *dispersion*, $H \geq 1$). Then:

$$Q^{\text{opt}} = q + \frac{B}{m}, \quad Q^{\text{ave}} = q + \frac{BH}{m}, \quad Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H-1)}{m}$$

- Equal weighting can never beat optimal in population.
- They coincide iff variances are homogeneous ($H = 1$).
- RSM with $k > 1$ pays off only when variances are **dispersed enough** (H large).

Average-subset RSM and the square-root rule

Proposition (Average-subset risk)

$$Q_k^{\text{sub}} = q + \frac{1}{k\bar{\tau}} + \frac{m-k}{k^2(m-1)} \frac{v_{\tau}}{\bar{\tau}^3} + O(k^{-3}) = q + \frac{B}{k} + O(k^{-2}).$$

Same shape as Q^{opt} , but divided by k instead of m — and the inflation cost is smaller, $I(T, k) < I(T, m)$.

Optimal k for average-subset RSM

$$L_k^{\text{sub}} \approx \left(\sigma_{\varepsilon}^2 + q + \frac{B}{k} \right) \frac{T-1}{T-k} \rightarrow \min \implies k_{\text{sub}}^* = \frac{-B + \sqrt{B^2 + (\sigma_{\varepsilon}^2 + q)BT}}{\sigma_{\varepsilon}^2 + q}.$$

$$k^* = \min\{m, \max\{1, \text{round}(k_{\text{sub}}^*)\}\} \sim O(\sqrt{T}).$$

Why the bagged optimum resists the same derivation

The $\sqrt{T}$ rule came from Taylor-expanding an *unconditional* expectation ($A_S = k\bar{\tau} + \epsilon_S$, $\mathbb{E}_S[\epsilon_S] = 0$, known variance):

$$Q_k^{\text{sub}} = q + \mathbb{E}_S\left[\frac{1}{A_S}\right] = q + \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + \dots$$

The bagged risk instead needs the *conditional* object

$$g_i(k) = k \mathbb{E}_S\left[\frac{1}{A_S} \mid i \in S\right] = k \mathbb{E}\left[\frac{1}{\tau_i + A_{S \setminus \{i\}}}\right].$$

- A **shifted conditional harmonic mean**: $A_S = \tau_i +$ (sum of $k - 1$ *other* precisions).
- Depends on the size- $(k - 1)$ precision sums of $\{\tau_j\}_{j \neq i}$ — a **different distribution for each i** .
- No closed form without restricting $\{\tau_i\}$: conditioning on $i \in S$ destroys the factorisation.

Way forward

Characterise R_k^{bag} by Monte Carlo — yet its *endpoints* are exact, enough to compare bagged RSM to both benchmarks with *no* functional form.

Infinite-bagged weights and the conservation law

Let $g_i(k) = k \mathbb{E}_S[\frac{1}{A_S} \mid i \in S]$. The infinite-bagged weight on forecast i is

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = \frac{k}{m} \mathbb{E}_S\left[\frac{\tau_i}{A_S} \mid i \in S\right] = \frac{\tau_i g_i(k)}{m}.$$

Since $\sum_{i \in S} w_{S,i} = 1$ for every S , we have $\sum_i \bar{v}_{k,i} = 1 \Rightarrow \sum_i \tau_i g_i(k) = m$ for all k .

Proposition (Conservation law)

For every $k \in \{1, \dots, m\}$,

$$\sum_{i=1}^m \tau_i \left[g_i(k) - \frac{1}{\tau_i} \right] = 0.$$

- $1/\tau_i$ is the g -reference giving *equal* weights: at $k = 1$, $g_i(1) = 1/\tau_i$ and $\bar{v}_{1,i} = 1/m$.
- Deviation $\tau_i[g_i(k) - 1/\tau_i] = m\bar{v}_{k,i} - 1$ is the *excess weight above equal weighting*, precision-rescaled.
- RSM redistributes weight relative to EW with **zero** net precision-weighted gain — holds for all k , all $\{\tau_i\}$, no distributional assumption.

Exact excess risk of the infinite-bagged RSM

Weight deviation $\Delta_{i,k} = \bar{v}_{k,i} - w_i^{\text{opt}}$ and excess risk $R_k^{\text{bag}} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}$.

Proposition (Exact excess risk)

Under common loading,

$$R_k^{\text{bag}} = \sum_{i=1}^m \frac{(\bar{v}_{k,i} - w_i^{\text{opt}})^2}{\tau_i} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2 \geq 0.$$

The common-loading term drops because $\iota'_m \Delta_k = 0$; only the idiosyncratic part $\sum_i \sigma_i^2 \Delta_{i,k}^2$ survives.

- **Endpoints (exact):** $R_k^{\text{bag}} = 0$ at $k = m$ ($\bar{v}_m = w^{\text{opt}}$); $R_1 = Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H-1)}{m} > 0$ unless all τ_i equal.
- **Sharp Jensen bound:** $R_k^{\text{bag}} \leq Q_k^{\text{sub}} - Q^{\text{opt}} = B z_k$, with $z_k = \frac{1}{k} - \frac{1}{m}$ — tight iff homogeneous.
- R_k^{bag} **strictly decreasing** in k — its decay pins down k^* .

Comparing bagged RSM to benchmarks — with *no* form

Write the inflated bagged loss using only its *endpoints* and sign:

$$L_k^{\text{bag},\infty} = (A + R_k^{\text{bag}}) \mathcal{I}(T, k), \quad A := \sigma_\varepsilon^2 + Q^{\text{opt}}, \quad \mathcal{I}(T, k) = \frac{T-1}{T-k}.$$

Structural facts only: $R_1 = \frac{B(H-1)}{m}$, $R_m = 0$, $R_k^{\text{bag}} \geq 0$, and $R_k^{\text{bag}} \downarrow$ in k .

Proposition (Bagged RSM vs. benchmarks, $1 \leq k < m < T$)

- (i) **Population bracketing:** $Q^{\text{opt}} \leq Q_k^{\text{bag},\infty} \leq Q^{\text{ave}}$ for every k .
- (ii) **Endpoint domination:** $L_1^{\text{bag},\infty} = L^{\text{ave}}$ and $L_m^{\text{bag},\infty} = L^{\text{opt}}$, hence

$$\min_{1 \leq k \leq m} L_k^{\text{bag},\infty} \leq \min(L^{\text{ave}}, L^{\text{opt}}).$$

(ii) is **unconditional** — it needs neither monotonicity nor a closed form, only that $k = 1$ and $k = m$ are admissible. RSM is therefore *never worse* than the better of the two benchmarks.

Exact comparison conditions (still no form)

Proposition (Continued)

(iii) Exact comparison conditions:

$$L_k^{\text{bag},\infty} < L^{\text{opt}} \iff R_k^{\text{bag}} < \frac{A(m-k)}{T-m},$$

$$L_k^{\text{bag},\infty} < L^{\text{ave}} \iff A(k-1) < R_1(T-k) - R_k^{\text{bag}}(T-1).$$

(iv) Interior strict domination: *the interior optimum beats both endpoints provided*

$$R_1(T-2) - R_2(T-1) > A \quad \text{and} \quad R_{m-1} < \frac{A}{T-m}.$$

Conservative (tractable) version

Jensen gives $R_k^{\text{bag}} \leq Bz_k$; substituting Bz_k for R_k^{bag} in (iii) yields *sufficient* conditions — exactly those from the average-subset $\sqrt{T}$ analysis.

Choosing k^* : when is it small, large, or interior?

Regime	Condition	k^*
Homogeneous forecasts	$H \approx 1$, so $R_1 \approx 0$	≈ 1
Long sample / small pool	$T \gg m$ (inflation negligible)	$\approx m$
Heterogeneous, moderate T	H large, $T \sim m$	interior $1 < k^* < m$
Endogenous / $\Sigma_{e,m}$ unknown	estimated forecasts	$\approx \lfloor \sqrt{T} \rfloor$

- **Homogeneous** $\Rightarrow$ gains vanish, RSM collapses to *equal weighting*.
- $T \gg m \Rightarrow$ estimation cheap, RSM tends to *full optimal*.
- **Dispersed errors, short T** $\Rightarrow$ an interior k^* strictly dominates both benchmarks (Prop., part iv).
- **Endogenous case** (forecasts themselves estimated): $k^* \approx \lfloor \sqrt{T} \rfloor$ is a robust *starting point*.

Bounding R_k from both sides: the optimal- k window

The Jensen ceiling $R_k^{\text{bag}} \leq Bz_k$ has a matching *projection floor*, giving an exact two-sided bracket:

$$\underbrace{\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2}_{\text{projection floor}} \lesssim R_k^{\text{bag}} \lesssim \underbrace{B z_k}_{\text{Jensen ceiling}}, \quad V_\delta = \frac{V_\tau}{\bar{\tau}^2}, \quad z_k = \frac{1}{k} - \frac{1}{m}.$$

(Floor: Cauchy–Schwarz projection of $\bar{v}_k - w^{\text{opt}}$ onto $w^{\text{ave}} - w^{\text{opt}}$; exact for ≤ 2 distinct precisions.)

A bounded window for k^*

$0 < z_k < 1 \Rightarrow z_k^2 < z_k$, so $R_k^{\text{bag}} \sim z_k^\alpha$ with $\alpha \in [1, 2]$, and the optimal subset size is **confined** to $\Theta(T^{1/3}) \leq k^* \leq \Theta(T^{1/2})$.

Cube root ($\alpha = 2$): bounded coherence *and* dispersed precisions. **Square root** ($\alpha = 1$): a dominant forecaster, or the endogenous case (estimated weights).

No functional form needed — Monte Carlo locates the optimum inside the window.

Summary of Section 4

- ① Unbiased combinations $\Rightarrow$ minimise variance $\sigma_\varepsilon^2 + w' \Sigma_{e,m} w$.
- ② RSM **unifies** equal weighting ($k = 1$) and optimal weighting ($k = m$); exact identities, no structure needed.
- ③ **Bagging bridge:** $Q_k^{\text{bag},\infty} \leq Q_k^{\text{bag},G} \leq Q_k^{\text{sub}}$, monotone in G ; with *no* structure, $Q^{\text{opt}} \leq Q_k^{\text{bag},\infty} \leq Q_k^{\text{sub}}$.
- ④ Under common loading: $Q(S) = q + 1/A_S$, the $Q^{\text{ave}} - Q^{\text{opt}} = B(H - 1)/m$ decomposition, and the **conservation law** $\sum_i \tau_i [g_i(k) - 1/\tau_i] = 0$.
- ⑤ **Exact excess risk** $R_k^{\text{bag}} = \sum_i (\bar{v}_{k,i} - w_i^{\text{opt}})^2 / \tau_i \geq 0$; square-root rule for average-subset k^* and a general FOC balancing gain vs. inflation cost.

Next: Monte Carlo evidence (Section 5).

Bates, J.M. & Granger, C.W.J. (1969). The combination of forecasts. *OR*.

Granger, C.W.J. & Ramanathan, R. (1984). Improved methods of combining forecasts. *J. Forecasting*.

Elliott, G. & Liao, Y. (2025). Forecast combination and the combination puzzle.

Elliott, G., Gargano, A. & Timmermann, A. (2013). Complete subset regressions. *J. Econometrics*.

Chen, J. & Maung, K. (2023). Nonparametric time-varying forecast combination.

Boot, T. & Nibbering, D. (2019). Forecasting using random subspace methods. *J. Econometrics*.